

Teachers

Dott. Tommaso Zoppi

Dott. Muhammad Atif

Lecture Schedule

Lectures will follow the schedule outlined below each week.

- Wednesday, 08:30-10:30 (9.15-10.30 til mid October)
- Friday, 14:00-17:00

Main Topics

- Metrology and Monitoring (Dott. Tommaso Zoppi)
- White/Black-box Testing - Fault/Error Injection (Dott. Tommaso Zoppi)
- Anomaly-Based Detection (Errors + Intrusions) (Dott. Muhammad Atif)
- Adversarial Attacks (for image-based classifiers) (Dott. Muhammad Atif)
- Robustness of (image-based) deep learners (Dott. Muhammad Atif)
- Deep Learning for Tabular data (Dott. Muhammad Atif)

Course Content

The course will expand on applications of Machine Learning (ML) techniques to support the design, monitoring and validation of cyber-physical systems that provide critical functionalities. Students will first be guided into the main attributes of critical systems, to introduce how ML algorithms or components may allow meeting security and/or safety requirements more efficiently or more reliably with respect to state-of-the-art techniques.

After introducing the need of integrating ML into critical systems, the course will move towards techniques to measure performance indicators of systems which become features of ML algorithms. Therefore, we will recap on metrology and we will elaborate on monitoring and data collection, presenting tools and best practices to extract knowledge from a (distributed) cyber-physical system. Monitoring activities typically provide data that has been and currently is used to support fault Injection and Robustness Testing Campaigns. Despite they do not heavily rely on ML, they usually complement monitoring activities and allow gathering labeled data to be used to train ML algorithms and therefore constitute an important pillar of this course.

Then, we will introduce Anomaly Detection and their role to perform either Error Detection or Fault Forecasting, which are important means to attain dependability. We will be starting from signature-based approaches, moving to supervised and [unsupervised ML](#) algorithms, highlighting their core mechanics, families and differences. Those algorithms are widely applied to detect intrusions and protect systems by implementing linear and non-linear classifiers. Therefore we will recap on common attacks and apply anomaly detectors to detect intrusions in public attack datasets.

In the last part of the course, the usage of deep neural networks (DNNs) in (safety-)critical systems will be explored, with a particular emphasis of the possible occurrence of failures and attacks. First, basic competences on DNNs will be acquired. Then, the robustness of image-based deep learners will be investigated, especially against the possible effects of RGB (Red-Green-Blue) camera failures. Next, adversarial attacks against image-based deep learners will be studied. Further, the possible effect of failures and attacks on graphical processing units (hardware accelerators for deep learning) will be discussed. Last, DNNs for tabular data will be presented, as they are an emerging solution also for anomaly detection and applicable to detect failures and attacks in supervised contexts.

The above arguments will be constantly applied by the Student during the course with several hands-on sessions.

Type of Assessment

The type of assessment is based on a project (mandatory) plus an interview.

ABOUT THE PROJECT: The project is related to one or more of the main topics of the course and it is assigned at the end of the lectures. Projects must be carried out individually.

The delivery of the project is done by Moodle and it will consist of i) a report and ii) ancillary work products.

The report should address at least the following points: i) description of the problem, ii) description of the applied methodology and the developed solution, and iii) description of the obtained results.

Work products consist in the software, configuration files, and all the files produced which may be necessary to entirely reproduced the activity.

ABOUT THE INTERVIEW: The oral interview has two targets: i) discussing some key element of the project, and ii) assessing the overall understanding of DCML topics by the student through questions about the course. The final mark will take into account both the project evaluation and the results of the oral interview.

The project must be delivered via the Moodle.

ADDITIONAL COMMENTS. Please remember the following:

- The mark assigned by the teacher is not "negotiable", meaning that the mark cannot be increased by taking an additional oral test or completing an extra assignment.
- If a student fails the exam, he/she cannot retake it before the next exam date.
- The students are strongly invited not to ask the teacher to apply exceptions to the above rules; in particular, they must avoid referring to their private or personal situation.

Teaching Material

- Slides presented during the course.
- Software, datasets and jupyter notebook made available during the course, through either github or google drive
- Papers published in Journals and in Conference proceedings.

 [Announcements](#) ○

 [Project \(26-11-2025\)](#) ○

 [Project Delivery - 1st Session \(due Jan 13th\)](#)
Due: Tuesday, 13 January 2026, 11:59 PM

 [Project Delivery - 2nd Session \(due Feb 3rd\)](#)
Due: Tuesday, 3 February 2026, 11:59 PM

▼ Introduction

The course will expand on applications of Machine Learning (ML) techniques to support the design, monitoring and validation of cyber-physical systems that provide critical functionalities. Students will first be guided into the main attributes of critical systems, to introduce how ML algorithms or components may allow meeting security and/or safety requirements more efficiently or more reliably with respect to state-of-the-art techniques.

 [Basics on Systems and Dependability - 24/09/2025](#) ○

▼ Metrology and Monitoring

After introducing the need of integrating ML into critical systems, the course will move towards techniques to measure performance indicators of systems which become features of ML algorithms. Therefore, we will recap on metrology and we will elaborate on monitoring and data collection, presenting tools and best practices to extract knowledge from a (distributed) cyber-physical system.

 [Metrology - 26/09/2025](#) ○

 [Monitoring \(Part I\) - 03/10/2025](#) ○

 [Monitoring \(Part I\) - 03/10/2025](#) ○

 [Monitoring \(Part II\) - 03/10/2025](#) ○

 [Testing - 15/10/2025](#) ○

✓ Lab Sessions

Miscellaneous Material related to Lab Sessions

 [Lab Session 08/10/2025: Git&Python](#) ○

 [More on GIT \(courtesy of prof. Ceccarelli - Software Engineering course\)](#) ○

 [Course GitHub](#)

✓ Data Analysis & Anomaly Detection

Then, we will introduce Anomaly Detection and their role to perform either Error Detection or Fault Forecasting, which are important means to attain dependability. We will be starting from signature-based approaches, moving to supervised and [unsupervised ML](#) algorithms, highlighting their core mechanics, families and differences. Those algorithms are widely applied to detect intrusions and protect systems by implementing linear and non-linear classifiers.

 [Data Analysis](#) ○

 [Supervised ML](#) ○

 [Unsupervised ML](#) ○

 [Meta Learning](#) ○

 [Supervised & Unsupervised ML \(Practice\)](#) ○

✓ Deep Learning for Critical Systems

Main arguments of this part:

- Basics of deep learning with Tensorflow.
- Robustness of deep learners. Robustness against image perturbation due to camera failures.
- Adversarial attacks and attack detection methods with the ART toolbox
- GPU faults and fault injection in deep learners
- Deep neural networks for tabular data.

To properly participate to the lectures, it is required to:

- regularly copy folders from ----- . Files are progressively updated and will be used during each lecture.
- have a google colab account <https://colab.research.google.com/>

 [DL_Lecture 1](#) ○

 [DL_Lecture2](#) ○

 [DL_Lecture3](#) ○

 [Camera_Failure_Lecture 2a](#) ○

 [Lecture2b_camera failure](#) ○

 [Lecture2c_CameraFailures](#) ○

 [Lecture3a_Adversarial Attacks](#) ○

 [Lecture3b_Adversarial Attacks](#) ○

 [Lecture3c_Adversarial Attacks](#) ○

 [Lecture3d_Adversarial Attacks](#) ○

 [Lecture 4a GPU Descriptionn](#) ○

 [Lecture 4b GPU Failure modes](#) ○

 [Lecture 4c](#) ○

 [Lecture 4d](#) ○

 [Links \(Deep Learning Lecture 1 & 2\)](#) ○

✓ **Description of the exam assignment** Not available

✓ **Seminars (13-12-2024)** Not available

